

Bloch Function and theorem:-

The main assumption of band theory is an electron in a metal is assumed to be moving in periodic potential due to +ve ion cores. The potential at 'r' distance from the +ve ion core is given by,

$$V(r) = \frac{q}{4\pi\epsilon_0 r} \quad \text{where } q \text{ is charge of nucleus}$$

of +ve ion core.

And hence the corresponding potential energy for electron is

$$U(r) = \frac{-eq}{4\pi\epsilon_0 r}$$

This energy for electron becomes infinity at close to atomic core as shown in fig.

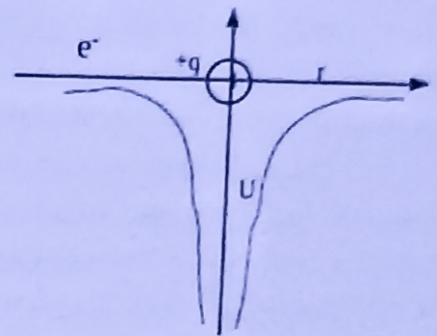


Fig: variation of potential energy for moving electron about +ve atomic core.

Bloch assumed that the electron inside metal are moving in periodic potential. The potential energy for moving electron at +ve ion site is infinity and is about zero exactly in between two +ve ion sites. This potential varies periodically with same period as the lattice i.e.

$$V(x) = V(x+a) \dots\dots\dots 1$$

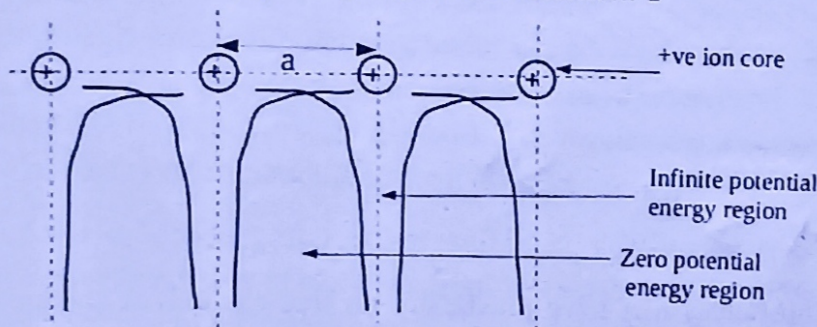


Fig: Periodic potential well in crystal lattice.

by Sir Sir

Hence, the corresponding 1-D Schrodinger wave equation for electron is written as,

$$\frac{d^2}{dx^2} \psi_k(x) + \frac{2m(E_n - V(x))}{\hbar^2} \psi_k(x) = 0 \dots\dots\dots 2$$

Bloch proved that an equation like 2 with periodic potential like in eqn 1 has a solution of form,

$$\psi_k(x) = e^{ikx} u_k(x) \dots\dots\dots 3 \quad \text{where } u_k(x) = u_k(x+a)$$

Regardless to this, Bloch gave a theorem called Bloch theorem which states that, "The eigen functions of wave equation for a periodic potential are the product of a plane wave e^{ikx} and a function $u_k(x)$ with periodicity of crystal lattice." eqn 3 represents Bloch function and Bloch theorem respectively.

Proof of Bloch theorem:-

Let us, consider 1-D linear chain of atoms of length 'L' containing N-atoms as shown in fig above. For an electron moving in such 1-D lattice, it suffers a periodic potential with periodicity equal to period of atomic lattice define as;

$$V(x) = V(x+a) \dots\dots\dots 1$$

Now, 1-D time independent Schrodinger wave eqn for an electron in such periodic potential is;

$$\frac{d^2}{dx^2} \psi_k(x) + \frac{2m(E_n - V(x))}{\hbar^2} \psi_k(x) = 0 \dots\dots\dots 2$$

The solution of above eqn according to Bloch is,

$$\psi_k(x) = e^{ikx} u_k(x) \quad \dots\dots\dots 3$$

where, $u_k(x)$ is a function representing the periodicity of lattice i.e.

$$u_k(x) = u_k(x+a) = u_k(x+Na) \quad .$$

The total length of lattice is expressed as; $L = Na$

Now, the wave function of electron at N^{th} atom potential well is given by;

$$\psi_k(x+Na) = e^{ik(x+Na)} u_k(x+Na)$$

$$\text{or, } \psi_k(x+Na) = e^{ik(x+Na)} u_k(x+Na)$$

$$\text{or, } \psi_k(x+Na) = e^{ikx} e^{ikNa} u_k(x) \quad [\text{being } u_k(x) = u_k(x+a)]$$

$$\text{or, } \psi_k(x+Na) = \psi_k(x) e^{ikNa} \quad [\text{being eqn 3}]$$

$$\text{or, } \psi_k(x+Na) = \psi_k(x) e^{ikL} \quad \dots\dots\dots 4$$

$$\text{or, } \psi_k(x+Na) = \psi_k(x) \cdot 1 \quad [\text{as } k = \frac{2N\pi}{L} \text{ for 1-D lattice and } e^{i2n\pi} = 1]$$

$$\psi_k(x+Na) = \psi_k(x)$$

This shows the wave function is same at 1^{st} and N^{th} atom potential with 'a' lattice periodicity. Further, from eqn 4, its complex conjugate from is;

$$\psi_k^*(x+a) = \psi_k^*(x) e^{-ikL} \quad \dots\dots\dots 5$$

Now; multiplying eqn 4 and 5, we get;

$$\psi_k(x+a) \psi_k^*(x+a) = \psi_k(x) \psi_k^*(x) e^{-ikL} e^{ikL}$$

$$\psi_k(x+a) \psi_k^*(x+a) = \psi_k(x) \psi_k^*(x) \quad \dots\dots\dots 6$$

Hence, this conforms the probability of finding the electron at N^{th} position ($x=L$) is equal to probability of finding electron at 1^{st} position ($x=0$). This is to say electron is uniform through out the crystal as in free electron theory. This proved Bloch theorem.

by S.G. Sir

#Effective mass of an electron:-

The electrons in a crystal are not completely free but interact with the periodic potential of crystal lattice hence their velocity is different in such a case compare to free electron and its mass gets changed. This mass of electron is called effective mass.

According to quantum mechanical theory, the velocity of an electron moving in periodic potential is equivalent to a wave packet (group of waves) and hence moving with velocity

' v_g ' define as, $v_g = \frac{d\omega}{dk}$

$$\text{or, } v_g = \frac{1}{\hbar} \frac{dE}{dk} \quad \dots\dots\dots 1 \quad [\text{being } \omega = \frac{E}{\hbar}]$$

For 'a' be the acceleration of particle then, Differentiating above equation w.r.to. 't', we get;

$$\frac{dv_g}{dt} = \frac{1}{\hbar} \frac{d \left[\frac{dE}{dk} \right]}{dt}$$

$$\text{or, } a = \frac{1}{\hbar} \frac{d \left[\frac{dE}{dk} \right]}{dk} \frac{dk}{dt}$$

$$\text{or, } a = \frac{1}{\hbar} \frac{d^2 E}{dk^2} \frac{dk}{dt} \quad \dots\dots\dots 2$$

An acceleration is developed on electron in lattice due to external applied 'E' electric field. So for m^* be its effective mass, then

$$F = m^* a \dots\dots\dots 3$$

Further, from newton's second law;

$$F = \frac{dp}{dt} = \hbar \frac{dk}{dt} \dots\dots\dots 4 \quad [\text{being } p = \hbar k]$$

from equation, 2, 3 and 4;

$$m^* a = \hbar \frac{dk}{dt}$$

$$\text{or, } m^* \frac{1}{\hbar} \frac{d^2 E}{dk^2} \frac{dk}{dt} = \hbar \frac{dk}{dt}$$

$$\text{or, } m^* \frac{1}{\hbar} \frac{d^2 E}{dk^2} = \hbar$$

$$\text{or, } m^* = \frac{\hbar^2}{\left(\frac{d^2 E}{dk^2} \right)}$$

$$\text{or, } m^* = \hbar^2 \left(\frac{d^2 E}{dk^2} \right)^{-1} \dots\dots\dots 5$$

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This is required expression for effective mass of electron in periodic crystal lattice. The variation of effective mass with wave vector 'k' is as shown in figure.

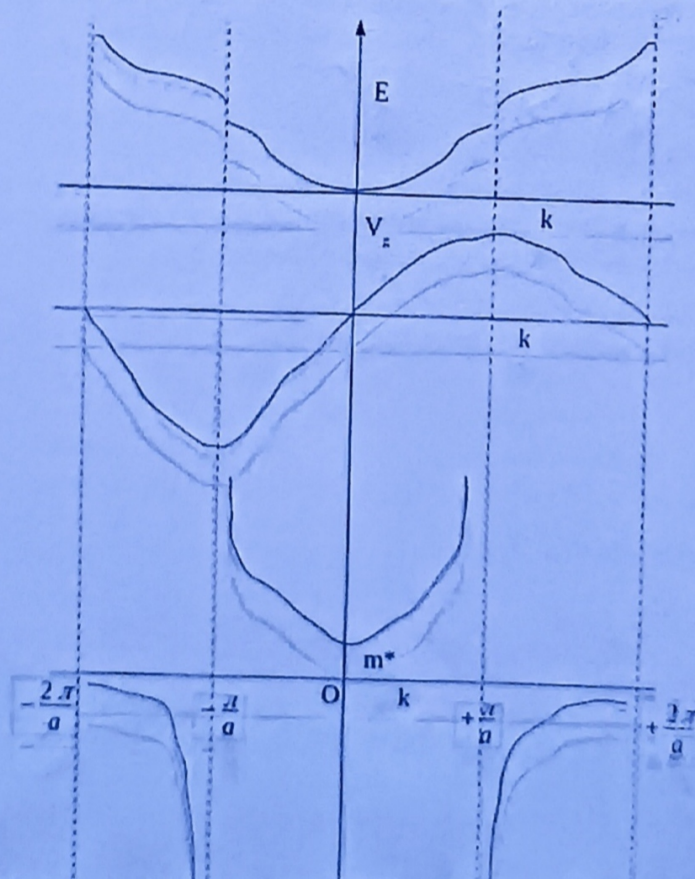


Fig: Variation of Energy , group velocity and effective mass of moving electron along with its momentum vector in a periodic lattice.

The mass of electron at $k=0$ is not equal to zero and this is equal to rest mass of free electron. The -ve value of effective mass refers to mass of holes. The effective mass value is +ve at bottom and is -ve at top as $\left(\frac{d^2 E}{dk^2}\right)$ have +ve value within $-\pi/2$ to $-k_0$, $+k_0$ to $+\pi/2$.

Tight Binding approximation:-

Tight binding approximation is an approach to calculate electronic band structure by linear combination of atomic orbitals or to say by the superposition of wave functions of an atom in crystal lattice site.

This approximation is quite good for valence bounded electron rather than conduction electron of an atom. The wave function for an electron or H-atom from solving radial form of Schrodinger wave function is;

$$\psi = A e^{(-r/a_0)}$$

This eigen function for two isolated atoms in one dimension is;

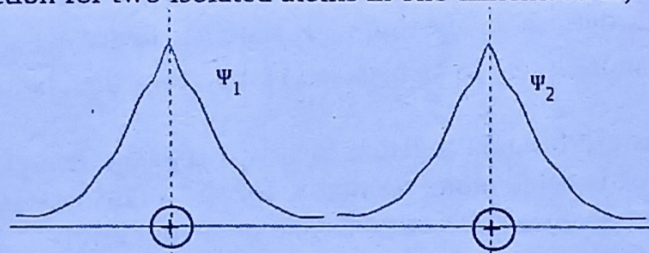


Fig: wave function about H- nuclei when two atoms are sufficiently far each other.

When these two atoms comes close together then atomic wave functions gets overlapped. For, two wave functions overlapping, if they are in phase this results to symmetric wave function and if they are out of phase this gives antisymmetric wave functions as shown in figure.

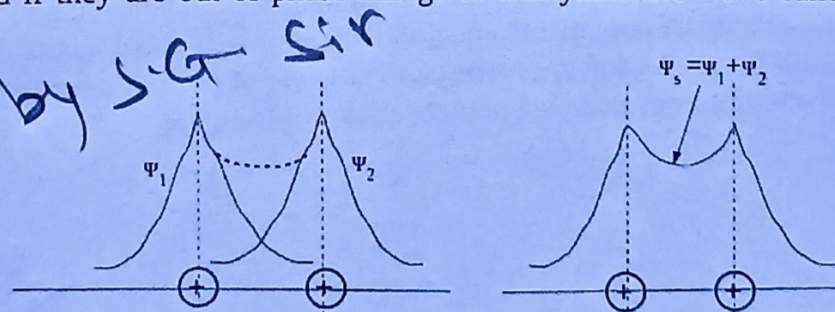


Fig: wave function overlap about H atom nuclei forming Symmetric wave function when two atoms are sufficiently close to each other.

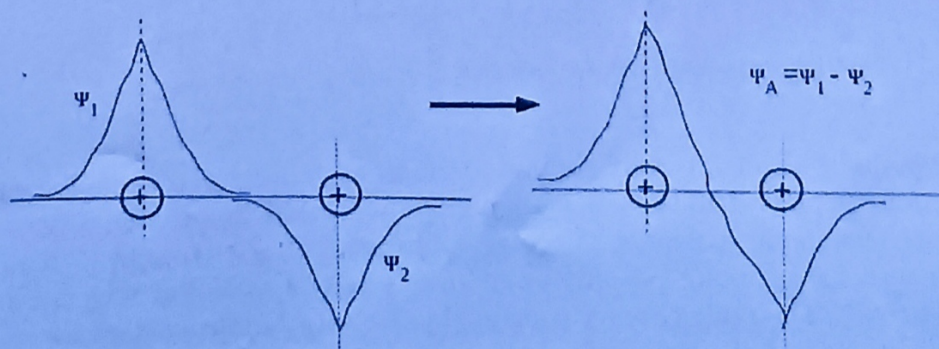


Fig: wave function overlap about H atom nuclei in out of phase forming AntiSymmetric wave function when two atoms are sufficiently close to each other.

Further, the probability distribution is given by;

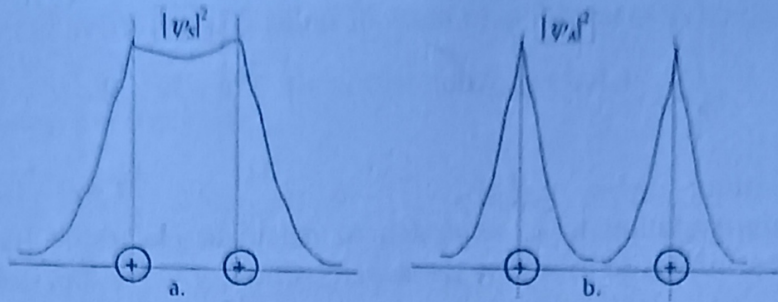


Fig: Probability density for a. Symmetric wave function and b. Anti-Symmetric wave function.

The probability of finding electron about mid point of two nuclei is symmetric in symmetric case and hence, probability for electron is same for either nuclei. Also, distribution value is greater than on either side of nuclei.

In case of anti-symmetric wave function, probability at mid point is zero and its value is maximum at the position of each nuclei. Further, as an electron in Ψ_s state has lower energy than one in Ψ_a . This is due to electron in Ψ_s is more likely to be between two protons than being at near to atomic ion.

Thus, when two atoms are brought together then two separate energy levels are formed from overlapping of each level of isolated atom. Similarly, for 'N' atoms are brought together this results 'N' discrete closely spaced levels and becomes a band of energy levels.

Kronig-Penny model: (motion of electron in Periodic potential)

This model is based on band theory and successfully describe the nature of material as conductor, semi conductor and insulator regarding their electrical conductivity. A real crystal lattice is an infinite array of lattice points with certain periodicity. Hence, the motion of an electron i.e. negative charge in crystal is through a periodic potential created due to positively ion core of atoms. The periodicity of potential is equal to that period of lattice points. The potential for an electron near to atom is zero and acts as free particle while at position between two atoms it is under a potential well and becomes bounded particle as shown in figure.

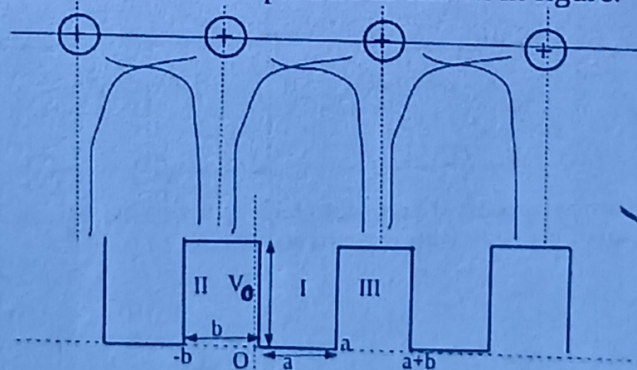


Fig: Square potential well approximated as in Kronig Penny Model.

As these periodic potential wells are very close to each other due to small order size of lattice and are parabolic in nature. To solve the Schrodinger wave equation for the wave function of an electron in such parabolic potential function is mathematically complex. Kronig-penny modification helps to solve Schrodinger wave equation more easily and hence the wave function of an electron in such periodic potential case is obtained.

Let, us consider a periodic square potential well of width 'b' i.e. barrier potential and of height V_0 and two barrier are separated by 'a' distance forming a well where potential is zero.

Thus, periodic potential is define as;

$$\begin{aligned} V(x) &= 0 & \text{for } 0 \leq x \leq a \\ &= V_0 & \text{for } -b \leq x \leq 0 \end{aligned}$$

Further for Ψ_1 , Ψ_2 and Ψ_3 be the wave function for the electron in I, II and III different regions as shown in figure.

The Schrodinger time independent wave equation in one dimension for I region is;

$$\frac{d^2}{dx^2} \psi_1(x) + \frac{2mE}{\hbar^2} \psi_1(x) = 0 \quad [\text{being } V=0 \text{ in first region}]$$

$$\text{or, } \frac{d^2}{dx^2} \psi_1(x) + \alpha^2 \psi_1(x) = 0 \quad \dots\dots\dots 1$$

$$\text{where, } \alpha^2 = \frac{2mE}{\hbar^2}$$

Similarly, The Schrodinger time independent wave equation in one dimension for II region is;

$$\frac{d^2}{dx^2} \psi_2(x) + \frac{2m(E - V_0)}{\hbar^2} \psi_2(x) = 0$$

$$\text{or, } \frac{d^2}{dx^2} \psi_2(x) - \frac{2m(V_0 - E)}{\hbar^2} \psi_2(x) = 0 \quad [\text{being } V > E \text{ in II region}]$$

$$\text{or, } \frac{d^2}{dx^2} \psi_2(x) - \beta^2 \psi_2(x) = 0 \quad \dots\dots\dots 2$$

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$$\text{where, } \beta^2 = \frac{2m(V_0 - E)}{\hbar^2}$$

The equation 1 and 2 is of second order differential equation form, whose solution is given by;

$$\psi_1(x) = A e^{i\alpha x} + B e^{-i\alpha x} \quad \dots\dots\dots 3$$

$$\psi_2(x) = C e^{\beta x} + D e^{-\beta x} \quad \dots\dots\dots 4$$

The wave function for the electron in III region can be written according to Bloch theorem as

$$\psi_3(x) = \psi_2(x) e^{ik(a+b)}$$

$$\text{or, } \psi_3(x) = (C e^{\beta x} + D e^{-\beta x}) e^{ik(a+b)} \quad \dots\dots\dots 5$$

Applying the boundary condition at $x=0$, we get;

$$\psi_1(0) = \psi_2(0)$$

$$\text{or, } A + B = C + D$$

$$\text{or, } A + B - C - D = 0 \quad \dots\dots\dots 6$$

Again,

$$\psi_1'(0) = \psi_2'(0)$$

$$\text{or, } A(i\alpha) + B(-i\alpha) = C(\beta) + D(-\beta)$$

$$\text{or, } (i\alpha)A + (-i\alpha)B + (-\beta)C + (\beta)D = 0 \quad \dots\dots\dots 7$$

Similarly, applying boundary condition at $x=a$, we get;

$$\psi_1(a) = \psi_3(a)$$

$$\text{or, } \psi_1(a) = \psi_3(-b) \quad [\text{being } \psi_1(a) = \psi_3(-b)]$$

$$\text{or, } e^{i\alpha a} A + e^{-i\alpha a} B = e^{-\beta b} e^{ik(a+b)} C + e^{\beta b} e^{ik(a+b)} D$$

$$\text{or, } e^{i\alpha a} A + e^{-i\alpha a} B - e^{-\beta b} e^{ik(a+b)} C - e^{\beta b} e^{ik(a+b)} D = 0 \quad \dots\dots\dots 8$$

Again, $\psi_1'(a) = \psi_3'(-b)$

$$\text{or, } i\alpha e^{i\alpha a} A + (-i\alpha) e^{-i\alpha a} B = (\beta) e^{-\beta b} e^{ik(a+b)} C - \beta e^{\beta b} e^{ik(a+b)} D$$

$$\text{or, } i\alpha e^{i\alpha a} A + (-i\alpha) e^{-i\alpha a} B + (-\beta) e^{-\beta b} e^{ik(a+b)} C + \beta e^{\beta b} e^{ik(a+b)} D = 0 \dots\dots\dots 9$$

Solving equations 6, 7, 8 and 9 by matrix method for non trivial solution;

$$\begin{vmatrix} A & B & C & D \\ 1 & 1 & -1 & -1 \\ i\alpha & -i\alpha & -\beta & \beta \\ e^{i\alpha a} & e^{-i\alpha a} & \beta e^{-\beta b} e^{ik(a+b)} & e^{\beta b} e^{ik(a+b)} \\ i\alpha e^{i\alpha a} & -i\alpha e^{-i\alpha a} & -\beta e^{-\beta b} e^{ik(a+b)} & \beta e^{\beta b} e^{ik(a+b)} \end{vmatrix} = 0$$

On simplifying above determinant, we get;

$$\frac{\beta^2 - \alpha^2}{2\alpha\beta} \sinh \beta b \sin \alpha a + \cosh \beta b \cos \alpha a = \cos k(a+b) \dots\dots\dots 10$$

This is the relation that is obtained by considering square potential well approximation. Sso, if it is replace by parabolic potential well as present in crystal then for this $\beta \gg \alpha$ and $b \rightarrow 0$ then $\sinh \beta b \rightarrow \beta b$, $\cosh \beta b = 1$

Hence eqn 10 for case of parabolic potential well becomes;

$$\frac{\beta^2}{2\alpha} b \sin \alpha a + \cos \alpha a = \cos ka$$

$$\text{or, } \frac{\beta^2 a b}{2\alpha a} \sin \alpha a + \cos \alpha a = \cos ka$$

$$\text{or, } \frac{p}{\alpha a} \sin \alpha a + \cos \alpha a = \cos ka \dots\dots\dots 11$$

where, $p = \frac{\beta^2 a b}{2}$ is a factor that define the attractive strength for an

moving electron by potential well. A larger value of 'p' signify increased value of potential barrier and electron is bounded to a particular well.

When $p \rightarrow 0$ then, from equation 11;

$$\frac{p}{\alpha a} \sin \alpha a + \cos \alpha a = \cos ka$$

$$\text{or, } \alpha a = ka$$

$$\text{or, } \alpha = k$$

$$\text{or, } \alpha^2 = k^2$$

$$\text{or, } \frac{2mE}{\hbar^2} = k^2$$

$$\text{or, } E = \frac{\hbar^2 k^2}{2m}$$

$$E = \frac{p^2}{2m}$$

This is the energy of an electron in term of momentum of free particle.

Here, energy of electron is purely equal to kinetic energy and particle is free to move in crystal. This case conforms that the solid is of conductor nature.

Again, for $p \rightarrow \infty$ then from equation 11;

$$\frac{1}{\alpha a} \sin \alpha a + \frac{\cos \alpha a}{p} = \frac{\cos ka}{p}$$

or, $\sin \alpha a = 0$

or, $\sin \alpha a = \sin n \pi$

or, $\alpha a = n \pi$

or, $\alpha = \frac{n \pi}{a}$

or, $\alpha^2 = \left(\frac{n \pi}{a}\right)^2$

or, $\frac{2 m E}{\hbar^2} = \left(\frac{n \pi}{a}\right)^2$

or, $E = \frac{\hbar^2 n^2 \pi^2}{2 m a^2}$

by SG Sir

This is the discrete energy levels in the case of particle bounded in one dimensional potential well and this indicate that the solid is of insulator nature.

Further, As $\cos ka$ has its values lies between +1 to -1. So,

$\left| \frac{p}{\alpha a} \sin \alpha a + \cos \alpha a \right| < 1$ this gives certain possible values of ' αa ' for the wave function to be Bloch function i.e. for a wave function representing periodic potential well.

The plot of $\left| \frac{p}{\alpha a} \sin \alpha a + \cos \alpha a \right|$ as a function of ' αa ' is as shown in fig.

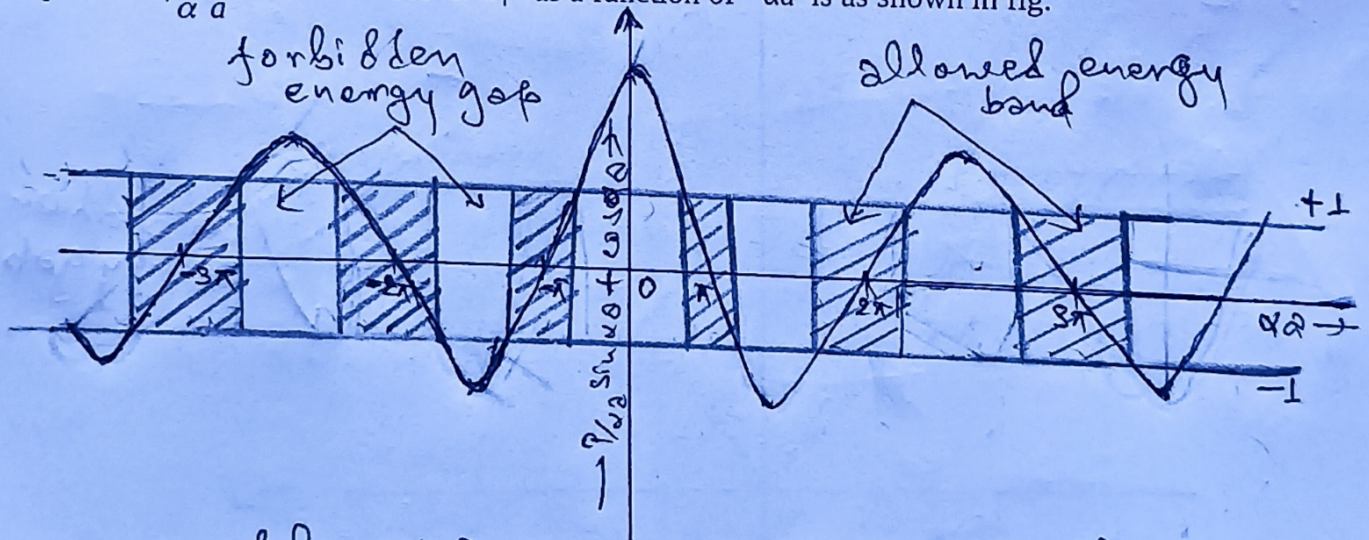


Fig-2 the plot of $\left| \frac{p}{\alpha a} \sin \alpha a + \cos \alpha a \right|$ as a function of αa .

On the basis of above plot, following conclusion can be drawn,

1. The energy spectrum consists of an infinite number of allowed energy bands. The gap between bands represents forbidden regions or bands. The boundaries of allowed energy levels is given by;

$$\cos ka = \pm 1$$

or, $\cos ka = \cos n \pi$

or, $ka = n \pi$

or, $k = \frac{n \pi}{a}$

2. As ' αa ' increases, the term $\frac{p}{\alpha a} \sin \alpha a$ decreases so that width of allowed energy band increases and hence, forbidden gap decreases and becomes narrow, representing conductor nature.
3. With increasing value of p , the width of allowed energy band decreases and at $p \rightarrow \infty$; the allowed energy band becomes infinitely narrow, representing insulator nature.

For $p \rightarrow 0$; the energy spectrum is quasi continuous.

For $p \rightarrow \infty$; the energy spectrum is a line spectrum.

The intermediate value of ' p ' refers to the position and width of the allowed energy band and hence the corresponding forbidden bands are obtained with vertical lines.

Further, the relation between the energy of electron and its wave vector is called dispersion relation. Thus, obtained curve with energy ' E ' with wave vector ' k ' is called dispersion curve.

Since,

$$k^2 = \frac{2mE}{\hbar^2}$$

by S.G. Sir

or, $E = \frac{\hbar^2 k^2}{2m}$ energy for a free particle and gives a parabolic curve.

As, for electron in periodic potential have discrete value of k i.e. $k = \frac{n\pi}{a}$, So the curves has discontinuities at points $k = \pm \frac{\pi}{a}, \pm \frac{2\pi}{a}, \pm \frac{3\pi}{a}, \dots$ as shown in figure.

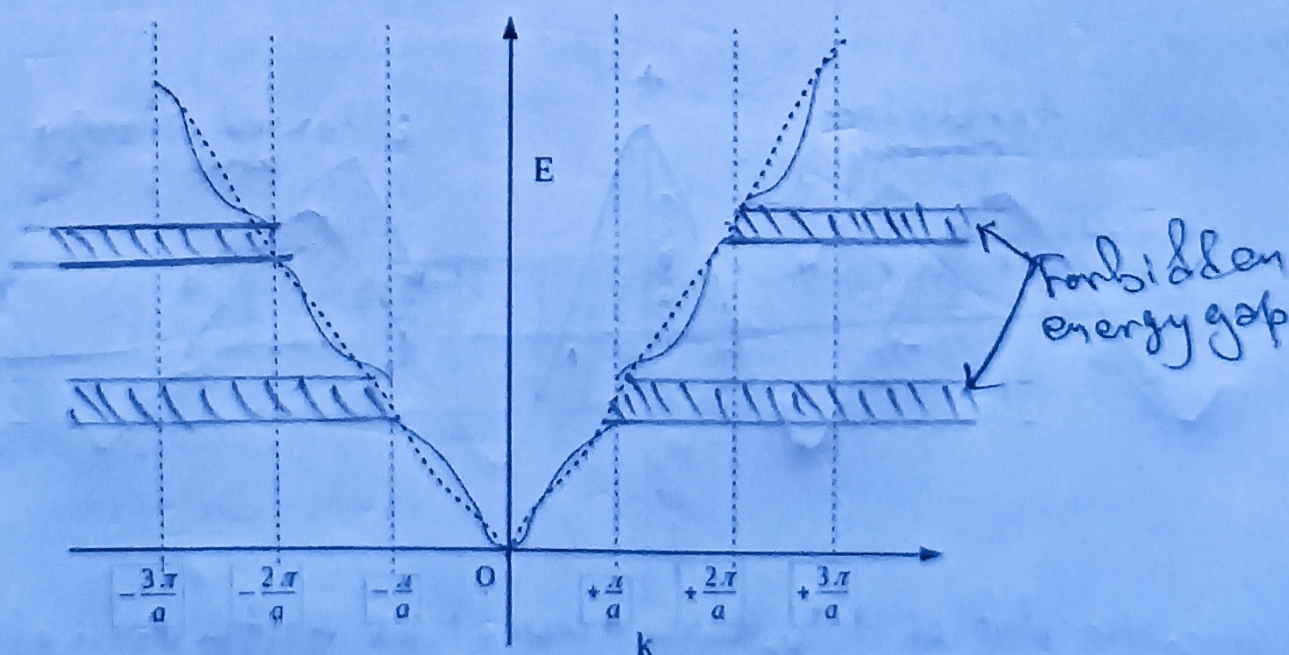


Fig: Dispersion relation for the electron moving in parabolic periodic lattice.

The region between $k = -\frac{\pi}{a}$ to $+\frac{\pi}{a}$ is called first Brillouin zone, Similarly from $+\frac{\pi}{a}$ to $+\frac{2\pi}{a}$ and $-\frac{\pi}{a}$ to $-\frac{2\pi}{a}$ represents for second Brillouin zone and on so on. The each line in curve represents band. The Brillouin zones are the space around the atom within which an electron have continuous particular momentum vector or energy levels.